



100% PRACTICAL SQL SERVER TRAINING

SQL Server & T-SQL Training

Build the database foundation, query fluency, and programming discipline required to work confidently with relational data.

RPA Vault Learning Series

SQL Server · T-SQL · Query Design · Database Programming

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THE COMPLETE SQL AND T-SQL LEARNING PATH

From your first SQL connection to reusable database logic.

The syllabus is structured as a practical progression. Each stage builds directly on the objects, statements, and query patterns introduced before it.

01 – Work inside SQL Server

Connect to Server; navigate SQL Server Management Studio, Object Explorer, Object Explorer Details, and Query Editor.

02 – Define reliable structures

Learn T-SQL foundations, DDL, database and table creation, alteration and deletion, data types, constraints, primary keys, and foreign keys.

03 – Create and retrieve records

Use INSERT, IDENTITY, INSERT...SELECT, UPDATE, DELETE, TRUNCATE, SELECT, WHERE, ORDER BY, DISTINCT, aliases, and common filters.

04 – Analyze relationships and results

Use functions, groups, HAVING, OVER/PARTITION BY, ranking, CTEs, TOP, operators, subqueries, set operators, and joins.

05 – Package database capability

Understand clustered/nonclustered indexes, views, cursors, stored procedures, user-defined functions, and triggers.

Skill stack

DESIGN

Database · Table · Types · Keys

MANIPULATE

Insert · Update · Delete · Truncate

QUERY

Select · Filter · Group · Join

PROGRAM

View · Procedure · Function · Trigger

Database objects

Database creation, alteration, deletion, tables, and data types.

Integrity rules

Constraints, primary keys, and foreign-key relationships.

Data changes

INSERT, IDENTITY, INSERT...SELECT, UPDATE, DELETE, and TRUNCATE.

Result filtering

SELECT, WHERE, BETWEEN, IN, LIKE, IS NULL, aliases, DISTINCT, and ORDER BY.

Analytical queries

Functions, GROUP BY, HAVING, window functions, ranking, CTEs, and TOP.

Reusable programming

Indexes, views, cursors, procedures, functions, and triggers.

Workspace

SSMS, Object Explorer, Query Editor

DDL

Databases, tables, types, constraints

DML & DQL

Insert, update, select, filters, sorting

Analysis

Functions, groups, windows, CTEs, joins

Programming

Indexes, views, cursors, procedures, UDFs, triggers

APPLIED LEARNING SEQUENCE:

Model

Create tables, choose data types, and enforce integrity with primary and foreign keys.

Populate

Insert and update records, then move data between tables through INSERT...SELECT.

Query

Filter, group, join, rank, and package results into reusable database components.

ENVIRONMENT, OBJECTS & DATA MANIPULATION

Build the structure before you query the data.

Establish the environment, database objects, data types, and integrity rules that make later SQL work reliable and understandable. This module covers both database design and maintenance statements.

SQL Server workspace

Connect to the server, navigate Object Explorer, inspect details, and use Query Editor to create, run, and refine T-SQL statements.

Database and table DDL

Create, alter, and delete databases and tables. Select data types that fit the information being stored.

Constraints and integrity

Use a primary key to identify a record and a foreign key to enforce the relationship between related tables.

Insert and identity

Add records with INSERT, use IDENTITY when an automatically generated value is appropriate, and create/populate a table from another table.

Change and remove rows

Use UPDATE for targeted modification. Compare DELETE and TRUNCATE so the scope and effect of a removal action is understood.

Move data between tables

Use INSERT...SELECT patterns to add rows from one table to another with a clear source and destination.

PRACTICAL BUILD: Create two related tables → define types, primary key, and foreign key → insert records → copy selected records to a second table → update one value → explain the difference between DELETE and TRUNCATE.

1. Design

Name tables and columns, assign data types, and identify the field that must stay unique.

2. Relate

Use a foreign key to connect the two tables and test the integrity rule with valid and invalid values.

3. Maintain

Populate rows, update a selected record, create a copy using INSERT...SELECT, and explain a safe deletion decision.

QUERY WITH PRECISION

Retrieve, refine, summarize, and compare.

This phase turns table data into useful information through filters, grouping, analytical functions, reusable query logic, and relational joins.

01 • SELECT & FILTER**Shape the right result set**

- SELECT and column aliases
- WHERE and SQL operators
- BETWEEN...AND, IN, LIKE, IS NULL
- DISTINCT and ORDER BY

02 • GROUP & ANALYZE**Find the pattern in the data**

- Aggregate functions
- GROUP BY and HAVING
- OVER (PARTITION BY ...)
- Ranking functions and TOP n

03 • REUSE LOGIC**Break a complex question into steps**

- Common Table Expressions
- Subqueries
- Operators inside expressions
- Intermediate query results

04 • CONNECT & COMPARE**Work across tables and result sets**

- INNER, OUTER, LEFT, RIGHT, FULL, CROSS joins
- UNION and UNION ALL
- INTERSECT and EXCEPT
- Join keys and null-aware output

PRACTICAL BUILD: Write a SELECT query → filter with WHERE, LIKE, and IS NULL → group and apply HAVING → rank rows within a partition → move the logic into a CTE → join a second table → compare results with UNION ALL or EXCEPT.

PERFORMANCE, REUSE & CONTROL

Move beyond one-off queries.

The final technical stage covers objects that support controlled data access, reusable logic, and well-organized database behavior.

Build a database layer that can be understood and reused.

Students learn the role of indexes, views, procedures, functions, cursors, and triggers—then connect those components into a clear database project narrative.

CAPSTONE DIRECTION

Create a small database; populate it; build joined reporting queries; add a view or stored procedure; and explain how the object design supports reliable access and maintenance.

Clustered & NonClustered Index

Understand table data organization vs separate query access paths.

Views

Create reusable query-based representations of data for simplified access or reporting.

Stored Procedures

Package repeatable database actions behind a named interface.

User Defined Functions

Create reusable logic that returns a single value or table result.

Cursors & Triggers

Understand row-by-row processing and event-driven behavior.

Read

Use indexes and well-structured queries to consider how data is accessed and retrieved.

Reuse

Use views, stored procedures, and user-defined functions to package logic for later use.

Respond

Use cursors and triggers with intent when row-by-row or event-driven behavior is required.

Design discipline

Keep object names, data relationships, and query intent clear for the next reviewer.

Performance awareness

Consider indexes alongside query design when choosing how data will be retrieved.

Reusable delivery

Use views, procedures, functions, and triggers to package logic intentionally.

PROJECT DELIVERY BRIEF:

Model the data

Create related tables and apply primary-key and foreign-key integrity.

Answer questions

Use joins, groups, rankings, or subqueries to produce useful output.

Package the logic

Add a view, stored procedure, or user-defined function and explain its purpose.

FROM SYLLABUS TO DEMONSTRABLE SKILL

What you will be ready to demonstrate.

The course connects essential database design to advanced analysis and database programming, producing a practical, explainable SQL project outcome.

Workspace confidence

Connect to SQL Server, use SQL Server Management Studio, navigate Object Explorer, and work productively in Query Editor.

Database design

Create databases and tables; choose data types; and apply primary/foreign-key constraints that protect relational integrity.

Query fluency

Write DML and DQL statements, manage data changes carefully, and create readable SELECT output with targeted filtering and sorting.

Analysis & programming

Use functions, groups, windows, CTEs, joins, indexes, views, procedures, functions, cursors, and triggers in a structured project context.

YOUR NEXT MOVE

Build a portfolio-ready SQL narrative.

Plan a relational database. Populate it with meaningful records. Query it to answer reporting questions. Package selected logic for reuse. Then explain the design and results clearly.

[Explore RPA Vault →](#)[Contact Us →](#)**1. Design**

Create a database, tables, types, and integrity rules that model a business situation.

2. Populate

Insert, update, copy, and maintain records while making deliberate change decisions.

3. Analyze

Filter, group, rank, join, and compare data to answer a focused reporting question.

4. Package

Use a view or programmable object to present a reusable database solution.

Ready to start?

Explore RPA Vault learning pathways or contact the team to discuss the SQL Server & T-SQL course.

[Explore RPA Vault →](#)[Contact RPA Vault →](#)

Build the language behind reliable data decisions.

Explore practical technology learning pathways and connect with RPA Vault to discuss the next step in your SQL Server and T-SQL journey.

Ready to build practical SQL skills?

Review RPA Vault learning options, discuss your course requirements, and choose a structured pathway for your technical goals.

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